PORTFOLIO CASE STUDY

From Manual Product Entry to a Governed Commerce System

Shopify Product Architecture | Metafield Taxonomy | AI-Assisted Content | Product Operations

ROLE: PRODUCT OWNER / DIGITAL & AI SYSTEMS

2022-2025

The operating problem

A luxury multi-brand catalog depended on inconsistent vendor data, repeated manual writing, fragmented image production, and product-by-product decisions. The catalog could continue growing; the operating model could not.

VARIED

INCONSISTENT INPUTS

Vendor files, line sheets, imagery, naming, and specifications followed no shared structure.

REPEAT

REPEATED DECISIONS

The same facts were researched and rewritten across PDPs, uploads, metadata, and campaigns.

SILOED

DISCONNECTED OUTPUTS

Content, imagery, collections, search, feeds, and reporting operated from separate interpretations.

The system mandate

1. **Normalize** — Standardize inputs
2. **Structure** — Define reusable fields
3. **Distribute** — Reuse verified facts
4. **Govern** — Add review + QA
5. **Scale** — Enable bulk workflows

CORE DESIGN QUESTION **How could verified product facts move from intake to publication without sacrificing accuracy, merchandising logic, or luxury brand standards?**

CONFIDENTIALITY NOTE

All product examples, interfaces, data tables, and workflow diagrams are reconstructed and anonymized. No client admin views, customer data, logos, proprietary spreadsheets, or third-party product photography are shown.

Translating business needs into a scalable product architecture

Automation was not the starting point. The first requirement was a shared definition of what the business needed to know about every product, how that information should be verified, and where it needed to appear.

CUSTOMER NEED

Understand the product quickly and confidently.

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FUNCTIONAL REQUIREMENT

Present consistent fit, material, silhouette, care, and construction information.

↓

SYSTEM RESPONSE

Capture facts in reusable fields instead of free-form descriptions alone.

MERCHANDISING NEED

Group products accurately by category, occasion, trend, and use.

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FUNCTIONAL REQUIREMENT

Create classification rules that do not depend on repeated individual judgment.

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SYSTEM RESPONSE

Define controlled attributes and collection logic.

OPERATIONAL NEED

Publish high product volume without duplicating research and writing.

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FUNCTIONAL REQUIREMENT

Create one source that supports PDPs, metadata, feeds, imagery, and campaigns.

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SYSTEM RESPONSE

Connect structured fields to downstream outputs.

CORE REQUIREMENTS

Accuracy · Consistency · Reusability · Flexibility · Governance · Scalability

SYSTEM DESIGN PRINCIPLE

Separate verified product truth from the way that truth is presented.

Vendor sources → Fact verification → Governed product record → PDPs · collections · search · SEO · alt text · feeds · campaigns · reporting

02 / PRODUCT INFORMATION ARCHITECTURE

Building one reliable source of product truth

Critical details were separated from free-form copy and stored as reusable attributes. Customer-facing language could evolve without changing the underlying product record.

CORE IDENTITY

Designer · title · type · style ID · season · color

MERCHANDISING

Occasion · trend · use case · seasonality · collection eligibility

GOVERNED PRODUCT RECORD

One source of verified facts

FIT & SILHOUETTE

Fit · length · rise · sleeve · neckline · sizing context

DISCOVERY

Search terms · SEO inputs · alt-text inputs · feed attributes

MATERIAL & CONSTRUCTION

Composition · trim · lining · texture · closure · care

DISTRIBUTION

PDP specs · bulk upload · campaign language · reporting inputs

ONE FACT, MULTIPLE USES

PDP CONTENT · COLLECTIONS · SEARCH · SEO · ALT TEXT · FEEDS · CAMPAIGNS

STRUCTURED PRODUCT TRUTH

Relaxed fit
70% viscose, 30% silk
Velvet trim
Sheer georgette sleeves
Hidden front placket

CUSTOMER-FACING PRESENTATION

A fluid silk-blend blouse with a relaxed silhouette, sheer sleeves, and velvet detailing. The wording can change by channel; the facts remain stable.

Structure the facts that must remain consistent. Preserve flexibility where human interpretation adds customer value.

Turning fragmented inputs into a repeatable publishing system

The workflow defined where automation could accelerate production, where human judgment was required, and which checks had to be completed before a product could go live.

1 Intake
Collect vendor files, line sheets, samples, imagery, and existing records.

2 Research
Resolve missing details through approved sources.

3 Verify
Classify facts as verified, observable, or unsupported.

4 Structure
Map facts into defined fields and controlled values.

5 Draft
Create titles, copy, metadata, alt text, and upload records.

6 Review
Check accuracy, tone, claims, search value, and accessibility.

7 Implement
Distribute approved data into Shopify, collections, and feeds.

8 QA
Validate product truth, imagery, mobile UX, variants, and discovery.

EVIDENCE CONFIDENCE

VERIFIED

Confirmed through vendor documentation, labels, samples, or approved sources.

OBSERVABLE

Clearly visible in imagery or the physical product but not formally documented.

UNSUPPORTED

Incomplete, contradictory, inferred, or excluded from publication.

AUTOMATION BOUNDARIES

AI + AUTOMATION ACCELERATED

Initial field mapping
Controlled-value recommendations
First-pass descriptions and metadata
Alt-text drafts
Bulk-file preparation
Repetitive transformations

HUMAN OWNERS RETAINED CONTROL

Source credibility
Fact verification
Fit and construction interpretation
Luxury editorial judgment
Image accuracy
Exceptions and final approval

Automation could accelerate a verified process. It could not replace the verification process.

Protecting product truth across every customer-facing output

Documented standards made quality transferable. Written and visual outputs were reviewed together because they had to describe the same product.

FACTS → PRESENTATION

FICTIONAL PRODUCT EXAMPLE

Governed facts

Relaxed silhouette
70% viscose, 30% silk
Velvet collar and trim
Sheer georgette sleeves
Hidden front placket
Removable neck tie

Customer-facing description

A fluid silk-blend blouse shaped with voluminous sheer sleeves and a relaxed silhouette. Velvet detailing defines the collar, while a removable tie offers another styling option.

IMAGERY STANDARDS

SILHOUETTE

Preserve real proportions, drape, volume, waist placement, and intentional asymmetry.

MATERIAL

Keep silk fluid, velvet dimensional, lace layered, and transparency believable.

CONSTRUCTION

Match seams, hems, closures, trim, panels, and front/back continuity.

PRESENTATION

Use clean backgrounds, soft frontal light, consistent crop, scale, naming, and image order.

Improve presentation without changing the product.

QUALITY-CONTROL LOOP

1. Standard defined →
2. Output produced →
3. Truth compared →
4. Issue classified →
5. Protocol refined →
6. Future output improved

SYSTEM IMPROVEMENT EXAMPLES

Inaccurate cuff → construction rule
Distorted volume → silhouette rule
Over-smoothed fabric → material rule
Inconsistent terminology → controlled vocabulary
Repeated metadata edits → field-definition update

Creating infrastructure that could scale beyond individual execution

The result was not simply faster entry. It was a governed commerce system connecting product truth, operating workflows, customer experience, merchandising, and future automation.

BEFORE

FRAGMENTED PRODUCTION

Inconsistent vendor inputs
Facts buried in emails and free text
Repeated research and rewriting
Informal content and image review
Individual interpretation drove classification
Automation risked scaling inconsistency

AFTER

GOVERNED PRODUCT SYSTEM

Verified facts captured once
Controlled attributes supported reuse
Content and imagery shared standards
Review responsibilities were explicit
Corrections became system rules
Automation operated within boundaries

BUSINESS VALUE CREATED

OPERATIONAL CONSISTENCY
A repeatable path across products, designers, and source formats.

CATALOG SCALABILITY
Fewer one-off decisions as new products entered the catalog.

CUSTOMER CLARITY
More consistent, scannable, and useful product information.

MERCHANDISING CONTROL
Reusable fields for collections, classification, search, and filters.

AUTOMATION READINESS
Structured inputs, rules, and review gates for responsible AI use.

PRODUCT-LEADERSHIP TAKEAWAY

The most valuable automation was designing a reliable system that determined what could be trusted, how information should move, and where human judgment remained essential.

MY OWNERSHIP

PRODUCT + SYSTEMS

Product strategy · requirements definition · information architecture · taxonomy and metafields · workflow design · AI governance · QA standards · documentation

CAPABILITIES DEMONSTRATED

Product ownership · Shopify operations · structured content · human-in-the-loop systems · cross-functional translation · change enablement